

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)		1 clockwise arrow (1) 1, 2, 3, 4 in correct positions 1 4 3 (1) 2		2		2		
	(b)		True because clockwise OR process 1 has greatest work than 3 (1) No work done for 2 & 4 (1) Apply ecf if needed			2	2		
	(c)		Attempt made at any area under the graph or enclosed or counting squares (1) 9.25ish large squares or 9×25 small squares or correct trapezium / enclosed area method (1) Answer = $4\,600 \pm 400$ [J] (1)			3	3	3	
	(d)		Heat flows from gas to surroundings (through piston) OR conduction through metal (1) Accept molecules collide with container [Thermal] equilibrium or same temperature as surroundings (1)		1	1	2		
	(e)		Any 4 × (1) from: • Volume increases [for expansion part] • Work done by gas [for expansion but not compression] • Internal energy decrease [for expansion but not compression] • Increased PE linked to lower temperature or expansion • KE or internal energy decrease linked to cooling or lower temperature or expansion • [Process] too quick for heat transfer			4	4		

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	(f)		Nitrogen boils OR heat transferred to LN ₂ OR high pressure inside bottle. Accept gas expands	1			1		
	(g)		Force on bottle / nozzle is equal and opposite to force on gas / nitrogen (1) Provides [anticlockwise] moment / torque [for rotation] (1) For 2 nd mark, do not accept “principle of moments” because this is relevant to equilibrium or adding moments. Don't accept equal and opposite forces acting on the walls	1	1		2		
	(h)		Sound [waves] produced (or implied) (1) Implication of pressure release / molecule or gas release / energy release OR displacement of particles/molecules OR vibration of particles / molecules (1)	1	1		2		
	(i)		Nitrogen escapes from the container OR nitrogen escapes into lift (1) ACCEPT pressure in lift builds up DON'T ACCEPT container explodes OR container acts like rocket Leading to suffocation / anything to do with breathing N ₂ (1)			2	2		
			Question 6 total	3	5	12	20	3	0